

Q-Observe the pre-configured MPLS network in the provided Packet Tracer or GNS3 file, and note down two routing behaviors that are different from standard IP routing.
Hint: Look for how labels are used and how packets move between routers.

Label Switching (Push, Swap, Pop): Unlike standard IP routing where routers re-evaluate the destination IP at every hop, the ingress router pushes a label onto the packet header. Intermediate routers (Provider/P routers) perform a simple swap of the label, and the egress router pops the label before passing the plain IP packet to its destination LAN.

Pre-Determined Label Switched Paths (LSP): Packets follow a fixed, predetermined path (LSP) through the network core based on label mappings established by signaling protocols like LDP (Label Distribution Protocol), rather than dynamically recalculating the best hop using standard IP routing protocols at every intermediate node.